Bradykinesia Analysis Workflow

Data Organization

All data must be stored in one folder. All files in the folder must be titled according to the folder name ("FolderName") with extensions describing the contents of each file. The contents of the folder are:

- Original video of the UPDRS examination titled "FolderName"
- MediaPipe output video titled "FolderName_Pose_MediaPipe.avi"
- MediaPipe body keypoint outputs tilted "FolderName_Body_Outputs.csv"
- MediaPIpe hand keypoint outputs titled "FolderName_Hand_Outputs.csv"
- OpenPose output video titled "FolderName_Pose_OpenPose.avi"
- Sub-folder titled "JSON_Files" containing each OpenPose output JSON file
- Spreadsheet titled "FolderName_Segmentation.xlsx" containing start and end frames of each bradykinesia assessment

Workflow

1. Run *PD_BradyKinesia_Analysis.m* in MATLAB; this function calls the subsequent functions in succession.
2. Sub-function *load_pose.m* prompts the user to select a folder. Videos and associated pose estimation data from the folder are loaded. Pose estimation data are segmented based on start and end frames from the Excel spreadsheet in the selected folder (specified in line 391).
3. Sub-function *gapFill_filter.m* automatically gap fills missing pose data using linear interpolation for gaps up to 0.12 s and applied a low-pass filter using a 4^{th} order Butterworth filter with a cut-off frequency of 5 Hz.
4. *findEvents.m* lets the user inspect and correct automatically detected events used to identify movement cycles of the bradykinesia tasks performed (Fig. 1). There are subplots corresponding to left and right limbs for each bradykinesia tasks. Use slider or use cursor to update the image to show the selected frame. To delete spurious events: use brush to select events and click 'Delete Events'. To create missing events: use cursor to pick frame with missing event and click 'Create Events'. To visually check if events have been selected correctly, click 'Summary Plot' which opens a new window with overlaid plots for each movements cycle for each separate bradykinesia task; the curves should follow a repeatable pattern. Click 'Save' when done.

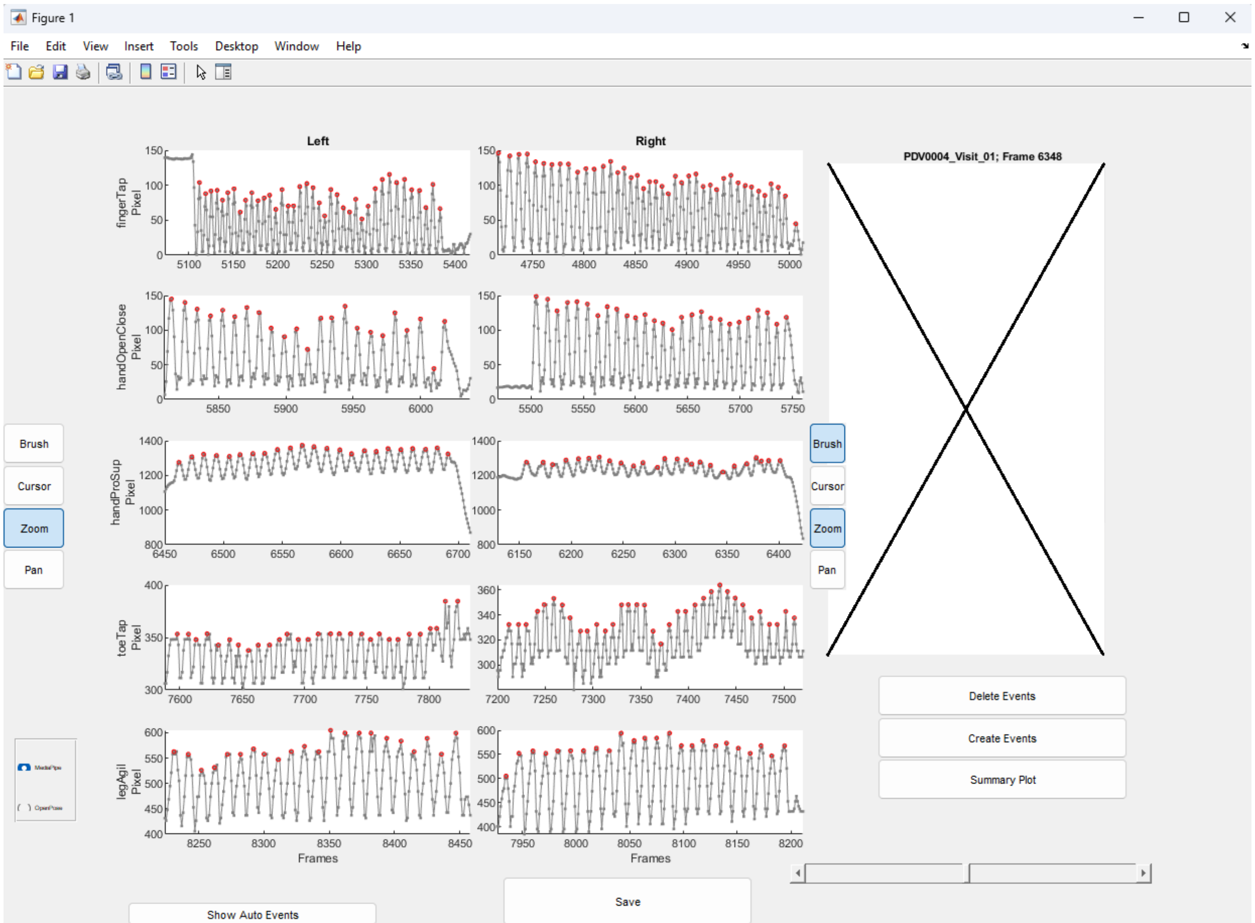

Figure 1. Graphical User Input to visually inspect and correct events of movement cycles.

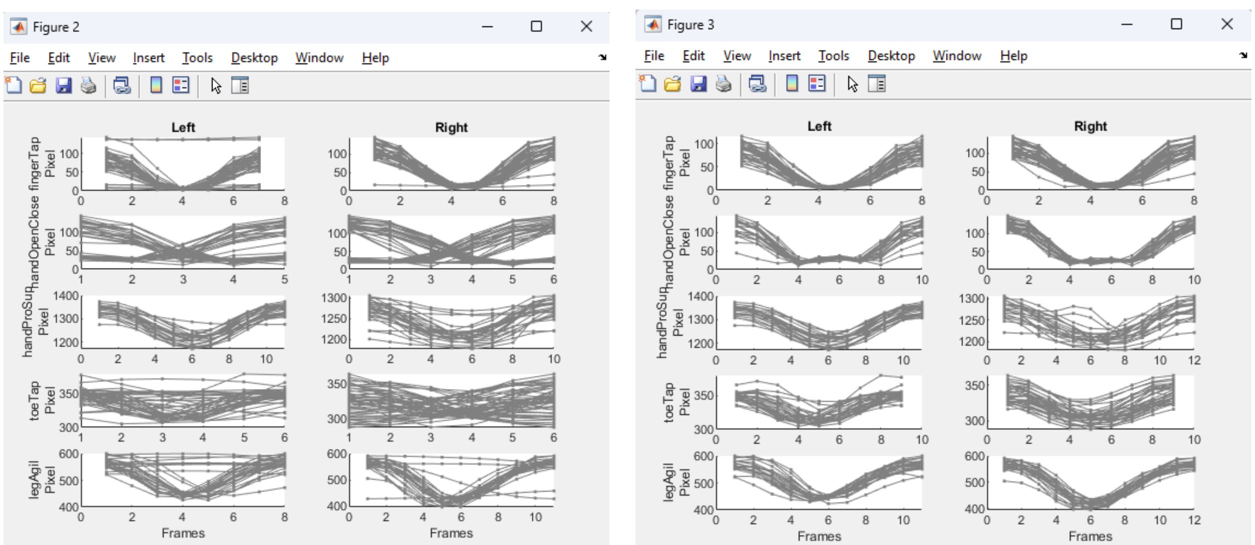

Figure 2. Summary plots to confirm whether events have been correctly selected. Left plot shows overlaid curves for each movement cycle *before* events have been correct; right plot shows curves *after* events have been corrected. Notice that curves on right plot are similar (with some degree of variability) while curves on left are more variable.

5. *calcMovementParameters.m* automatically calculates temporal and spatial movement parameters.
6. Contents saved in MATLAB data file
   The saved '.mat' file contains 10 structures: 'amplitudeParameters', 'auto_events', 'bradyKin_analyze', 'bradyKin_segments', 'data', 'frameInfo', 'output_name', 'temporalParameters', 'timeStamp' and 'videoInfo'.

   amplitudeParameters
   Struct with amplitude variables for bradykinesia tasks (excluding hand pronation/supination).

   auto_events
   Struct with frame numbers of events for all bradykinesia tasks. Note that frame numbers refer to each segments (i.e., a frame number of 10 refers to the tenth frame within a specific segment).

   bradyKin_analyze
   Struct with logical variables. Variables are true by default. Variables can manually be set to false if only a portion of the bradykinesia tasks are performed.

   bradyKin_segments
   Struct with start and end frames of all bradykinesia tasks. Frame numbers are imported from the Excel spreadsheet containing start and end frames.

   data
   Struct containing pose data for OpenPose or MediaPipe.

   frameInfo
   Struct containing logical vectors indicating whether the pose data have been gap filled.

   output_name
   Name of selected folder.

   temporalParameters
   Struct with temporal variables for bradykinesia tasks.

   timeStamp
   Vector containing time stamps for each frame in original video.

videoInfo
Struct containing VideoReader variables for original video and output videos for OpenPose and MediaPipe. The VideoReader variables contain meta-information of videos (e.g., frame rate and pixel resolution).